

Digital Logic Hardware Implementation

GATE EC 2014 Circuit Analysis

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PROBLEM STATEMENT

Q37. Consider the circuit shown in Fig. 3.20. If $C = 0$ the expression for Y is

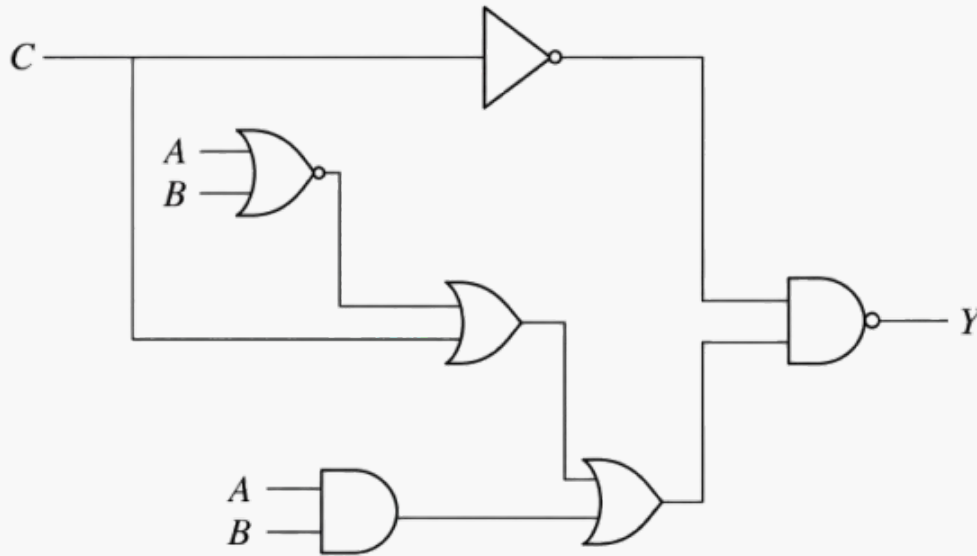


Fig. 3.20

- a) $A\bar{B} + \bar{A}B$
- b) $A + B$
- c) $\bar{A} + \bar{B}$
- d) AB

TRUTH TABLE ANALYSIS

A	B	$X = \overline{A + B}$	$Q = AB$	$R = X + Q$	$Y = \bar{R}$
0	0	1	0	1	0
0	1	0	0	0	1
1	0	0	0	0	1
1	1	0	1	1	0

STEP-BY-STEP DERIVATION

Step 1: Evaluate the Upper NOR Gate

The upper-left gate is a NOR gate with inputs A and B .

$$X = \overline{A + B}$$

Applying De Morgan's theorem,

$$X = \bar{A}\bar{B}$$

Thus, the output of the NOR gate is

$$\boxed{X = \bar{A}\bar{B}}$$

Step 2: Evaluate the First OR Gate

The output of the NOR gate is ORed with the input C .

$$P = X + C$$

Substituting $X = \bar{A}\bar{B}$,

$$P = \bar{A}\bar{B} + C$$

Since the problem specifies

$$C = 0$$

therefore

$$P = \bar{A}\bar{B}$$

Step 3: Evaluate the Lower AND Gate

The lower gate is an AND gate with inputs A and B .

$$Q = AB$$

Thus, the output of the AND gate is

$$\boxed{Q = AB}$$

Step 4: Evaluate the Second OR Gate

The outputs P and Q are applied to the next OR gate.

$$R = P + Q$$

Substituting the obtained expressions,

$$R = \bar{A}\bar{B} + AB$$

Step 5: Evaluate the Inverter Branch

Input C is also connected to a NOT gate.
Since

$$C = 0$$

the inverter output becomes

$$\bar{C} = 1$$

Step 6: Evaluate the Final NAND Gate

The final gate is a NAND gate whose inputs are R and $\bar{C}$.

$$Y = \overline{R \cdot \bar{C}}$$

Substituting $\bar{C} = 1$,

$$Y = \overline{R \cdot 1}$$

$$Y = \bar{R}$$

Substituting the value of R ,

$$Y = \overline{\bar{A}\bar{B} + AB}$$

Applying De Morgan's theorem,

$$Y = (\bar{A}\bar{B})'(AB)'$$

$$Y = (A + B)(\bar{A} + \bar{B})$$

Expanding,

$$Y = A\bar{B} + \bar{A}B$$

Hence, the given circuit realizes the Exclusive-OR (XOR) function.

$$Y = A\bar{B} + \bar{A}B$$

Correct Option : (a)

$$A\bar{B} + \bar{A}B$$

HARDWARE IMPLEMENTATION

1. Components Required

- 1x Arduino UNO
- 1x Breadboard
- 2x Push Buttons (Inputs A and B)
- 2x 10k Ω Pull-down Resistors
- 1x LED
- 1x 220 Ω Resistor
- Jumper Wires

2. Hardware Realization Using Arduino

The simplified Boolean expression obtained from the circuit is

$$Y = A\bar{B} + \bar{A}B$$

which represents an XOR operation.

Input Connections

- Push Button A $\rightarrow$ Arduino Digital Pin 2
- Push Button B $\rightarrow$ Arduino Digital Pin 3
- One terminal of each push button $\rightarrow$ +5V
- 10k Ω resistor from each input pin to GND

LED Output Connections

- Arduino Digital Pin 4 $\rightarrow$ 220 Ω Resistor
- 220 Ω Resistor $\rightarrow$ LED Anode (+)
- LED Cathode (-) $\rightarrow$ GND
- Arduino GND $\rightarrow$ Breadboard GND Rail
- Arduino 5V $\rightarrow$ Breadboard Power Rail

Working Principle

The Arduino reads the logic states of inputs A and B and evaluates

$$Y = A\bar{B} + \bar{A}B$$

using software.

The computed output Y is transmitted to Arduino Digital Pin 4, which drives the LED.

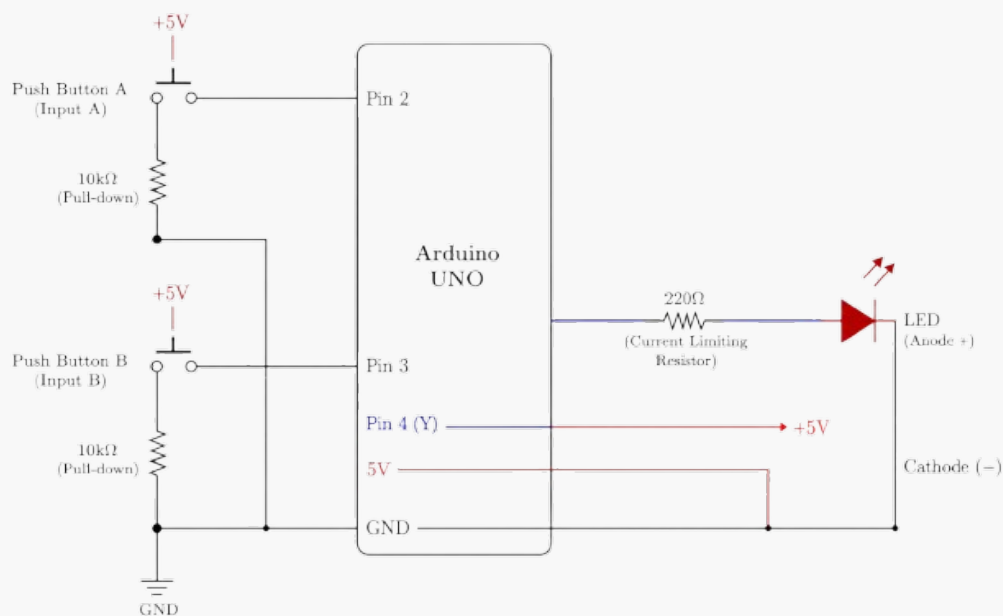
- When $Y = 0$, the LED remains **OFF**.
- When $Y = 1$, the LED turns **ON**.

Thus, the hardware implementation verifies the XOR operation using an LED indicator.

Truth Table Verification

A	B	Y	LED
0	0	0	OFF
0	1	1	ON
1	0	1	ON
1	1	0	OFF

HARDWARE BLOCK DIAGRAM



Overview:

This block diagram illustrates the hardware implementation of the XOR function

$$Y = A\bar{B} + \bar{A}B$$

using an Arduino UNO, two push buttons and an LED.

Push Button A and Push Button B are connected to Arduino Pins 2 and 3 respectively. The Arduino evaluates the XOR expression in software and generates the output Y on Pin 4.

The output is connected through a 220Ω current-limiting resistor to an LED. The LED glows when $Y = 1$ and remains OFF when $Y = 0$, thereby verifying the XOR operation.

SOURCE CODE REPOSITORIES

The source code used for implementing the XOR function

$$Y = A\bar{B} + \bar{A}B$$

using Arduino UNO and LED-based hardware verification is available below.

Arduino C++ Program

[XOR_logic.ino](#)

Assembly Program

[XOR_logic.asm](#)

Project Repository

[Gate-Hardware-Logic Repository](#)

FINAL RESULT

Final Result

The given logic circuit simplifies to the Exclusive-OR (XOR) function.

$$Y = A\bar{B} + \bar{A}B$$

In the hardware implementation, the Arduino evaluates the XOR logic and drives an LED connected to the output pin.

- LED OFF $\Rightarrow Y = 0$
- LED ON $\Rightarrow Y = 1$

Hence, the correct answer is

$$Y = A\bar{B} + \bar{A}B$$

Correct Option: (a)